

# **The Digestive System**

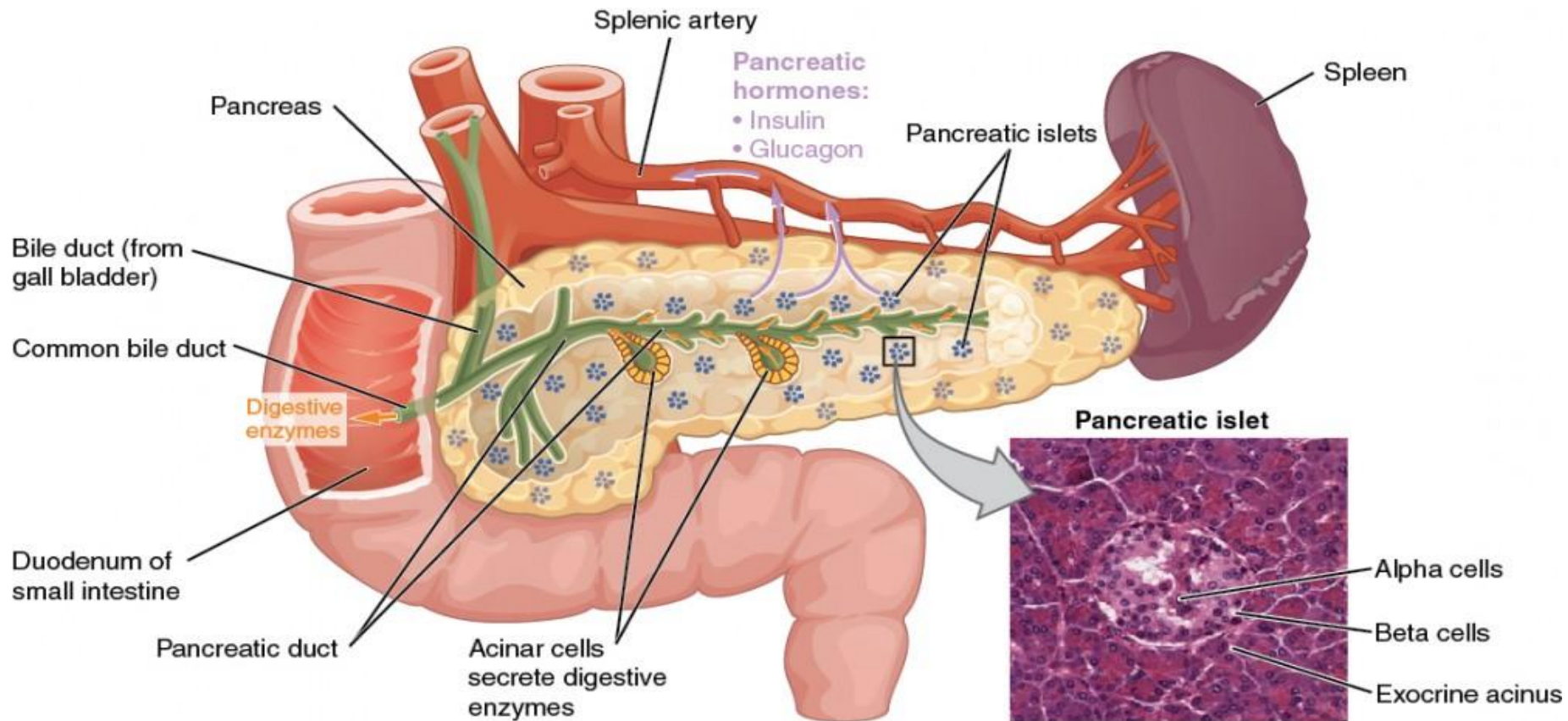
## **Histological structures of Pancreas**

**LG3**

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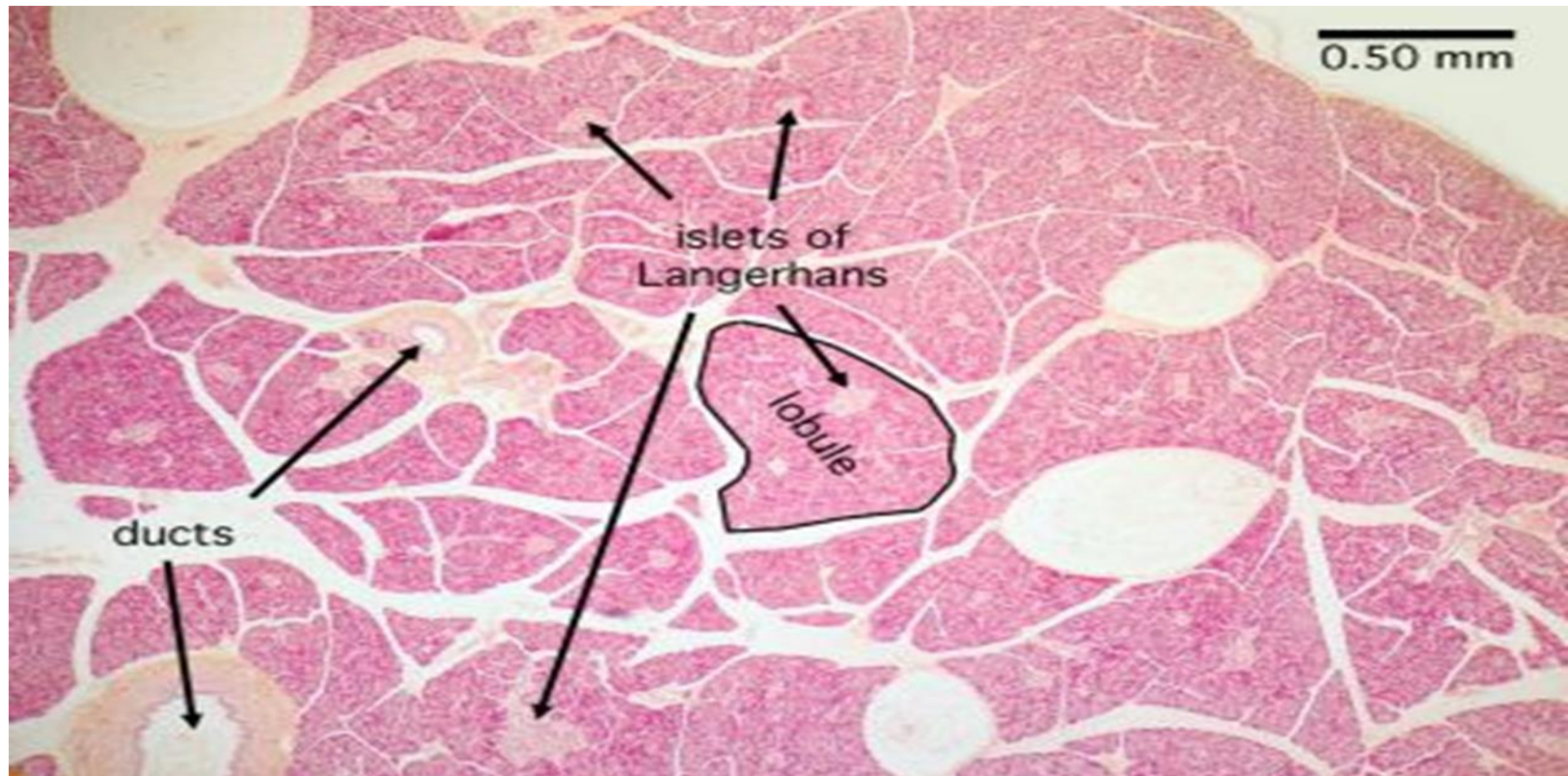
**Anatomy and Pathology department**

- **Pancreas** : It is an elongated retroperitoneal organ, with a large head near the duodenum and more narrow body and tail regions that extend to the left
- Is mixed **exocrine & endocrine** gland that produce digestive enzymes & hormones . The enzymes are stored & released by the cells of the exocrine portion **the exocrine portion consist of cells & ducts**
- The hormones are synthesized in islets of Langerhans



# The exocrine part of Pancreas:

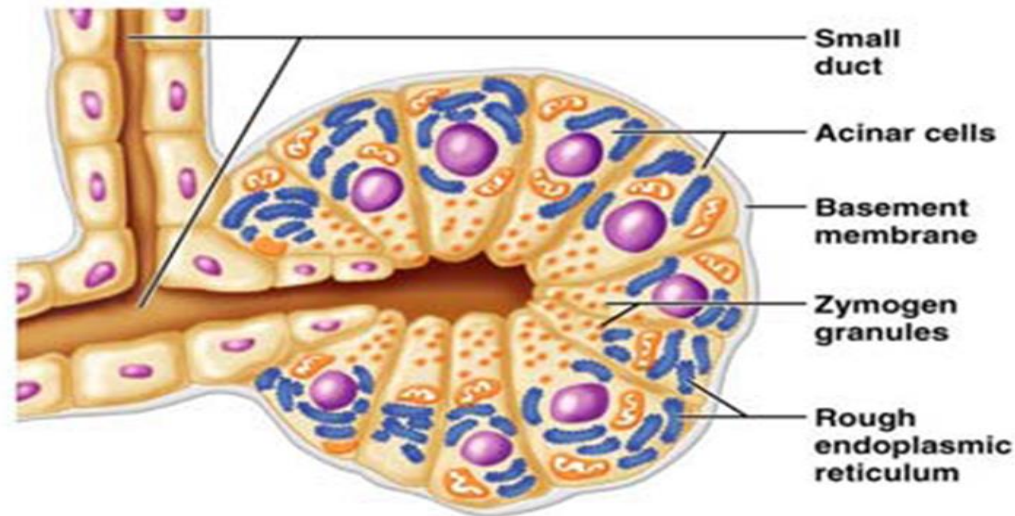
- A thin layer of loose c.t. surround the gland, the septa extend from it divided into lobules & lobes that carry b.v., nerve....
- Is compound acinar gland
- It secretes enzymes that play a very important role in the digestion of carbohydrates, proteins & fats



# Cell types in exocrine portion

**1-pancreatic acinar cells** : each acinus consist of several pyramidal – shaped enzyme secreting cells

- The basal part of cell is basophilic cytoplasm contain RER, ribosomes with secretory or zymogen granules in the cytoplasm mainly in the apical part of the cell. These granules are eosinophilic•
- The enzymes packaged & stored in membrane bound Zymogen granules



(a)

These enzymes controlled by cholecystikinin produced by enteroendocrine cells in duodenal mucosa



# **exocrine function**

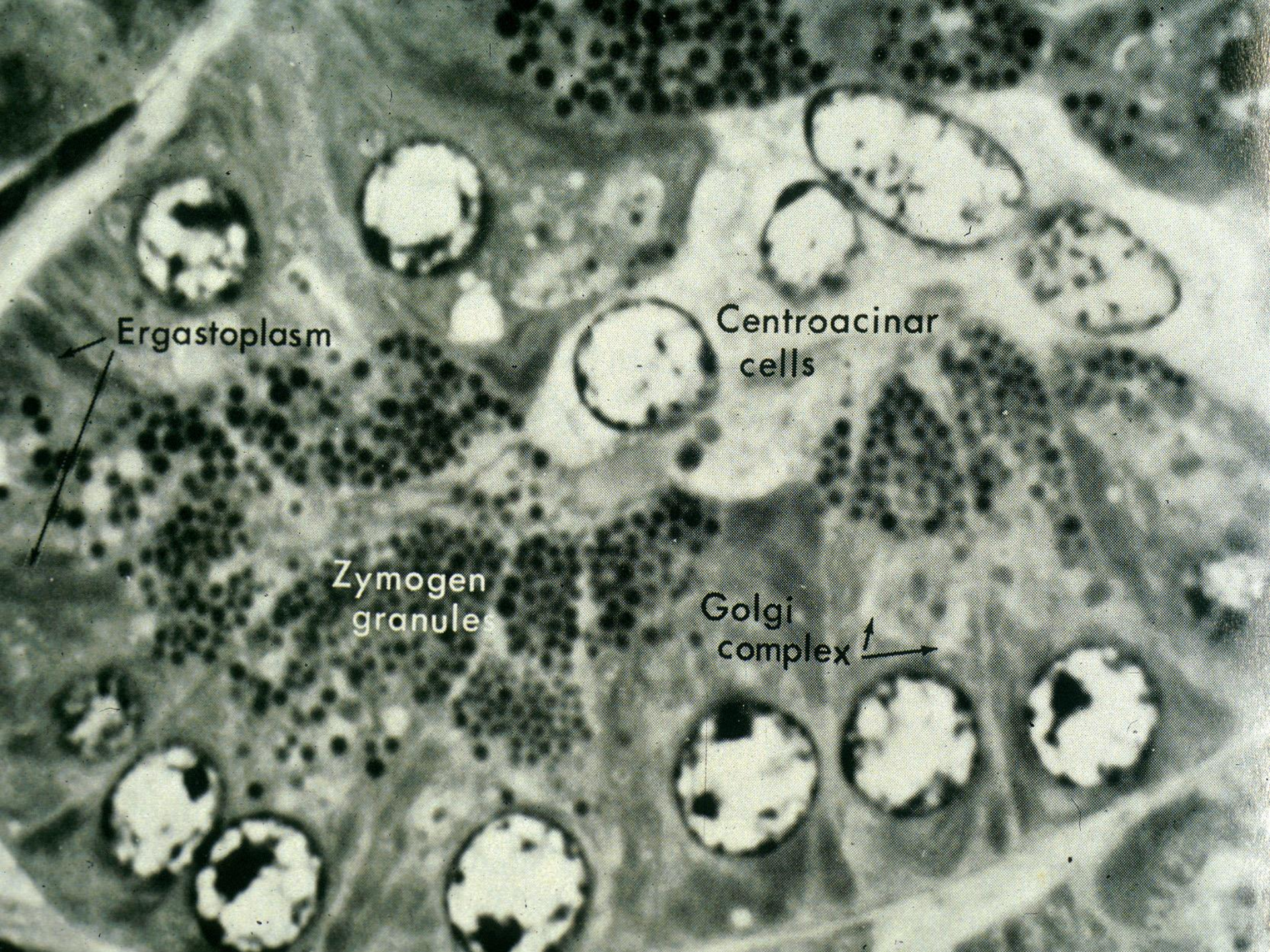
**Secrete following enzymes for breaking down carbohydrates, proteins, and fats.:**

- **Amylase → carbohydrates**
- **Lipase & related enzymes(Phospholipase A2 and Cholesterol esterase → fats**
- **Proteases (trypsin, chymotrypsin, carboxypeptidase, elastase) → proteins**



Acinar cells





Ergastoplasm

Centroacinar  
cells

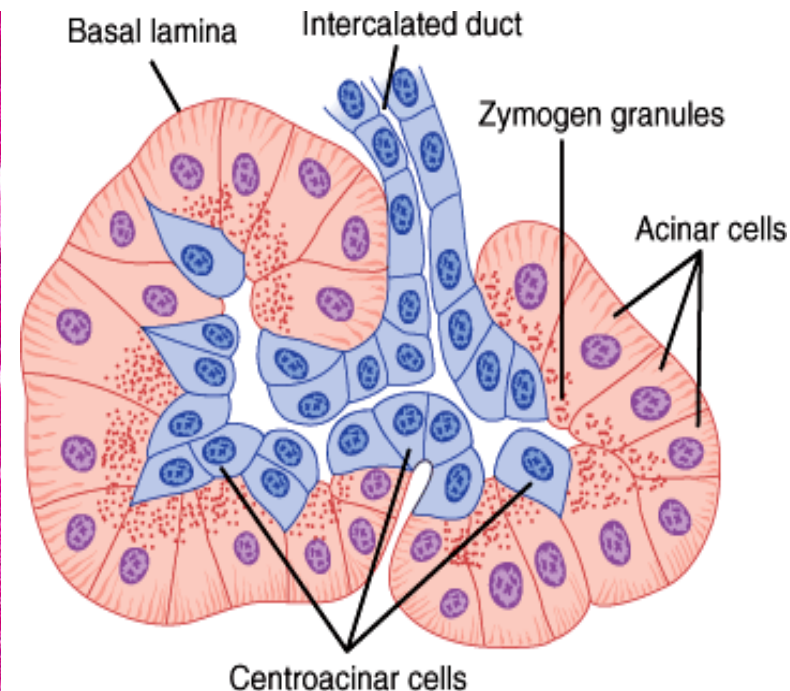
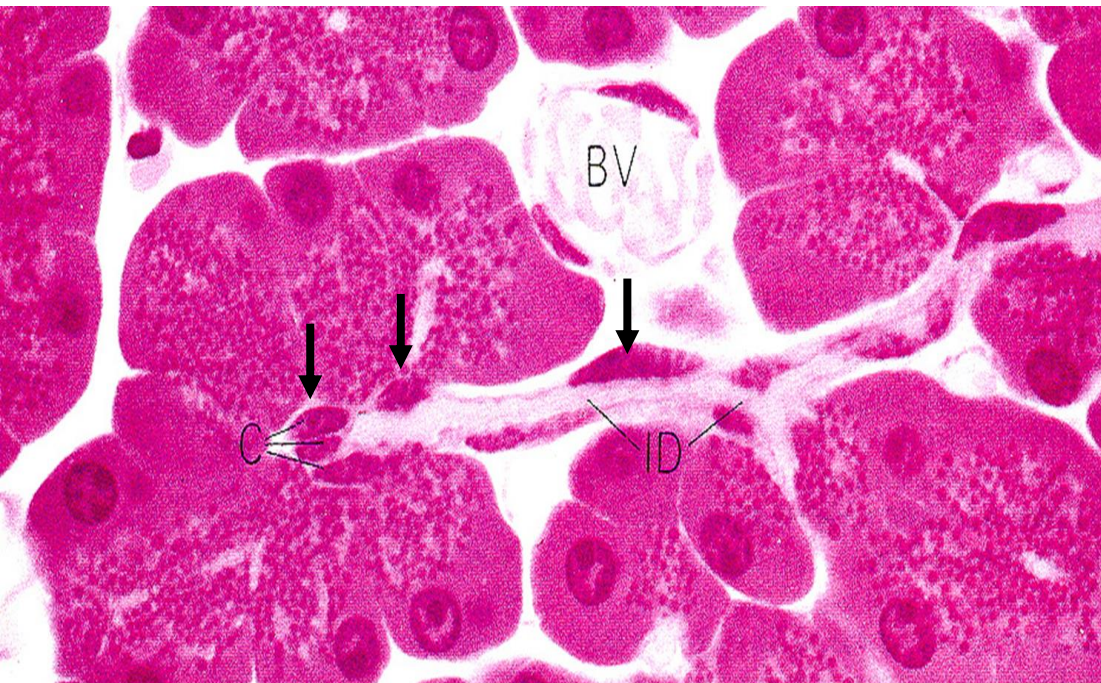
Zymogen  
granules

Golgi  
complex



# 2-centroacinar cells

- Each cell has condensed nucleus, clear cytoplasm
- 1-2 can be found in the lumen of each acinus at the origin of the intercalated duct
- Cells produce a **watery bicarbonate rich fluid** in response to stimulation by hormone **secretin** secreted by enteroendocrine cells in small intestine
- This fluid helps to adjust the acidic chyme to neutral pH to optimize pancreatic enzyme activity





- **Duct system:**
- The duct system of the pancreas is the network of channels that collects and transports pancreatic secretions (enzymes + bicarbonate) from the glandular acini to the duodenum. It plays a central role in digestion.
- **1- intercalated duct :**
- smallest ducts.
- Begin directly from the centroacinar cells inside the pancreatic acini.
- Lined by simple squamous or low cuboidal epithelium
- **2-intralobular**
- Formed by the joining of multiple intercalated ducts within a single lobule.
- Lined by low columnar epithelium.
- Carry secretions further toward larger ducts
- **3- inter lobular** duct found between lobules in septa lined by columnar epithelium may contain goblet cells& surrounded by large amount of connective tissue
- **4. Main pancreatic duct** :Formed by convergence of many interlobular ducts . Runs the length of the pancreas from tail → head . Joins the common bile duct to open into the major duodenal papilla (ampulla of Vater).Regulated by the sphincter of Oddi.
- **Acinar cells → intercalated ducts → intralobular ducts → interlobular ducts → main pancreatic duct → duodenum.**





Intercalated duct



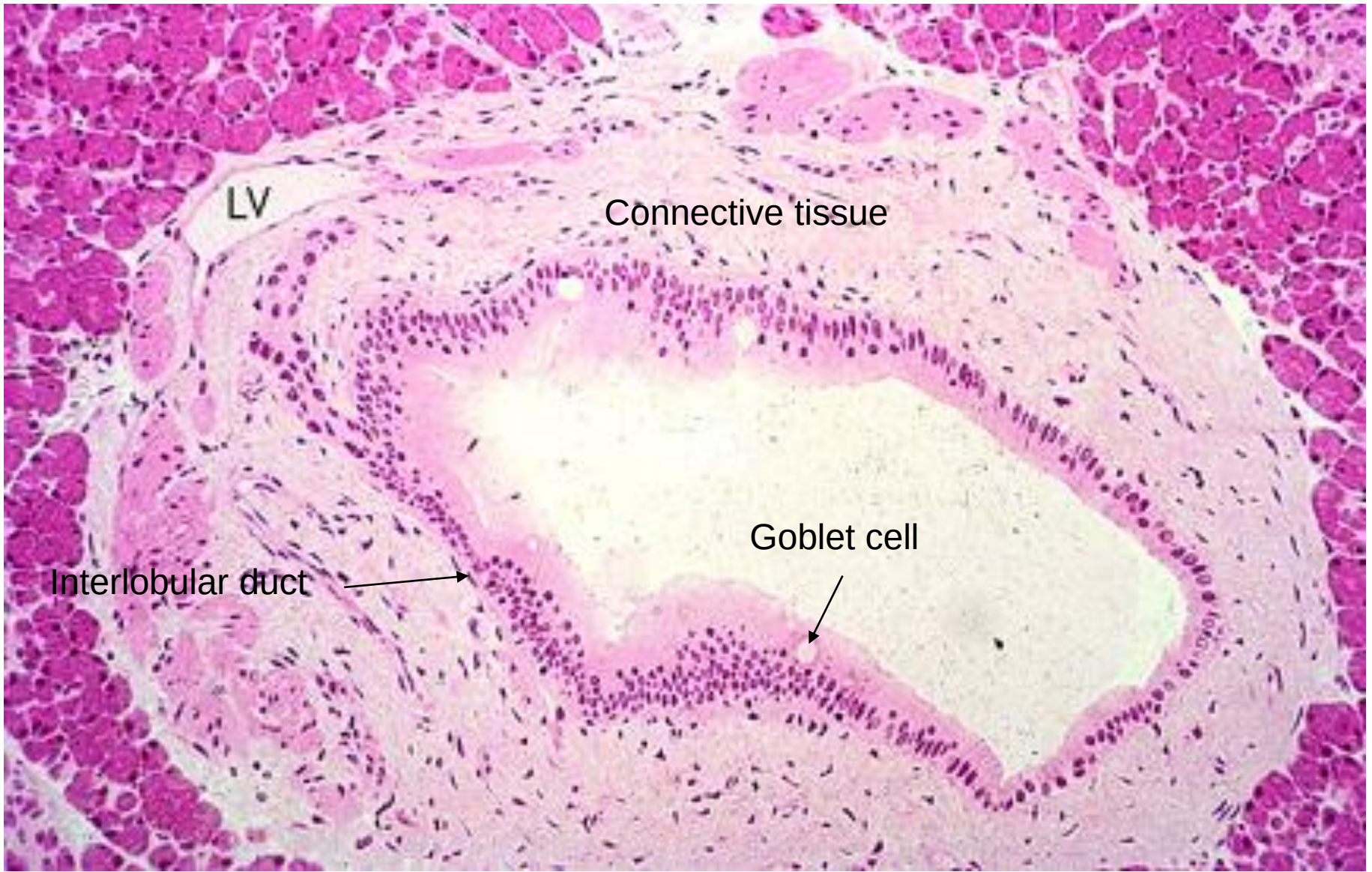


Intralobular duct

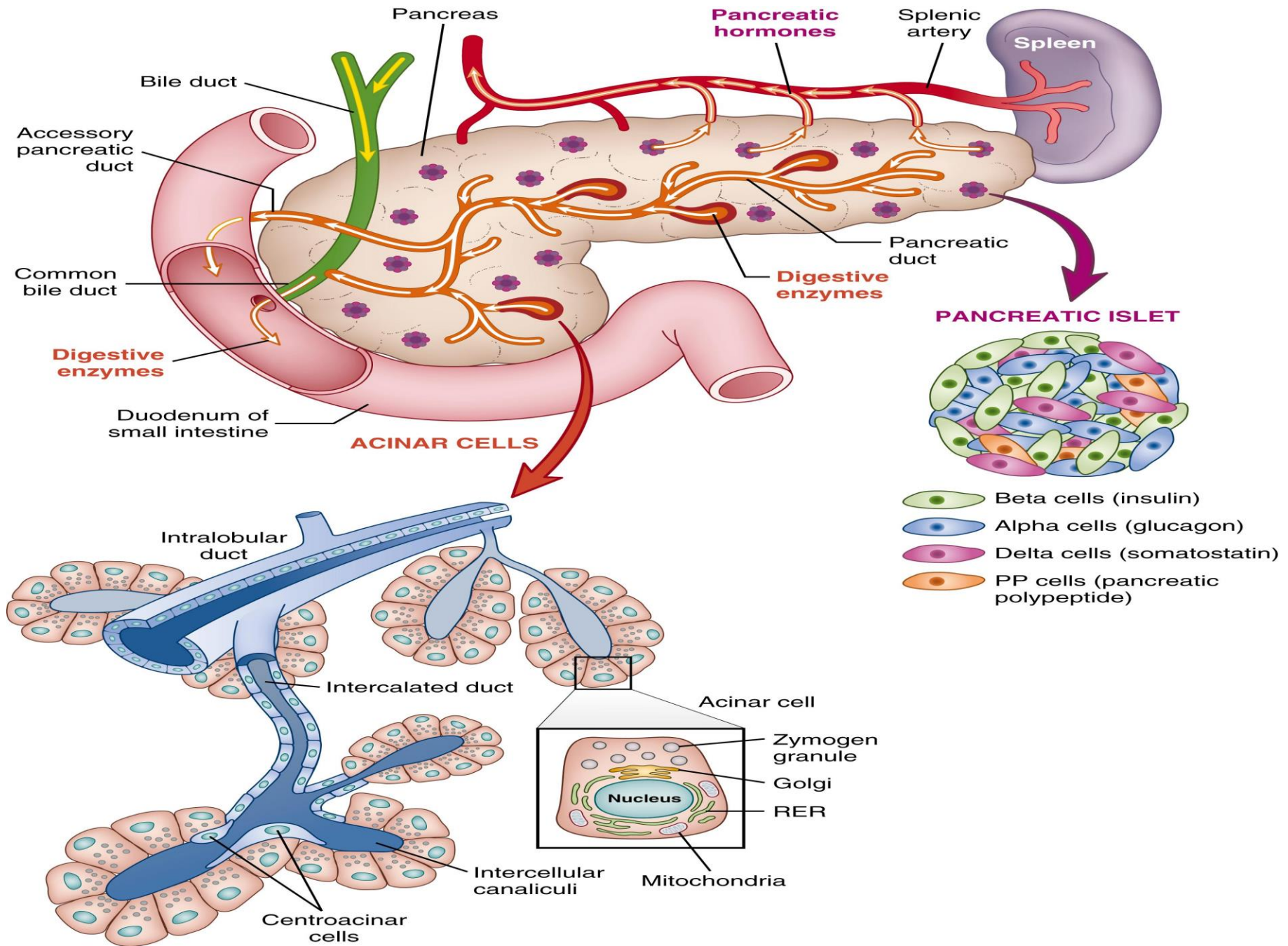
Connective tissue

Exocrine cells





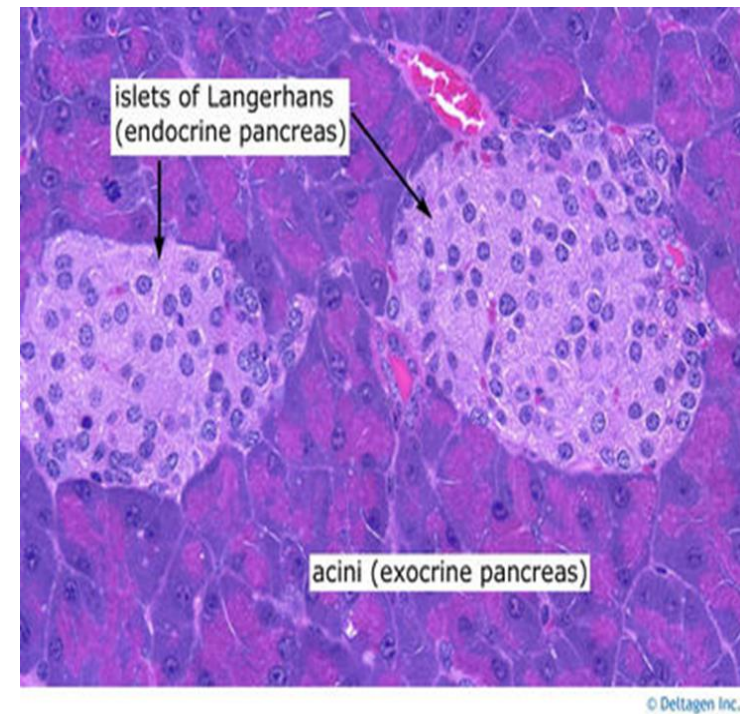






# Endocrine portion = Islets of Langerhans

- Are cluster of rounded cells , most numerous in tail
- Are smaller , paler than exocrine cells , polygonal in shape the cytoplasm contain secretory granules contain hormones, these cells are separated by network of fenestrated capillaries



Each islet surrounded by reticular fibers of c.t.

There are 4 major types of cells located in islets :A-B-D-F

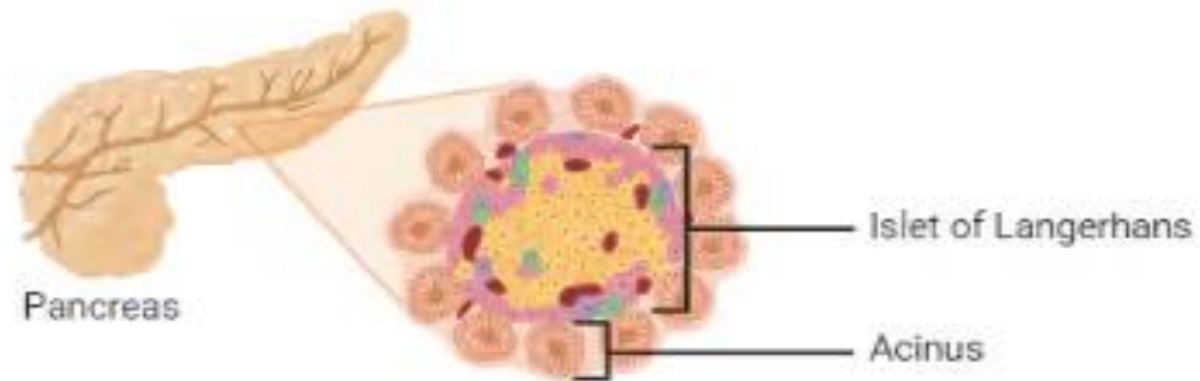
1.A cells(alpha)-periphery-secrete=**glucagon**-20%, increases blood glucose

2.B cells(beta)-center –secrete=**insulin**-%70, decreases blood glucose

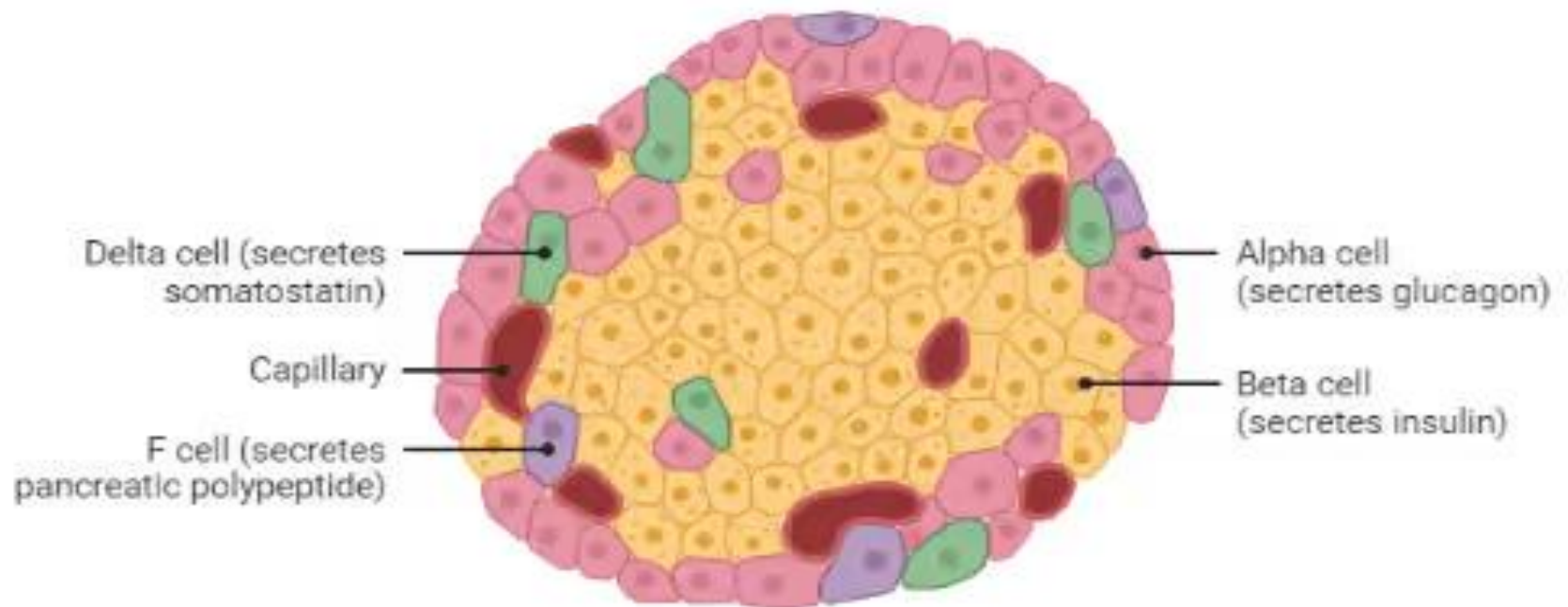
3.D cells(delta)- variable -secrete = **somatostatin** .  
inhibits the secretion of glucagon & insulin.

4- PP cells(F-cells)-variable-secrete **pancreatic polypeptide cells**,, their secretion inhibit pancreatic exocrine secretion of enzymes & bicarbonate. and It also cause gall bladder relaxation & decreases bile secretion

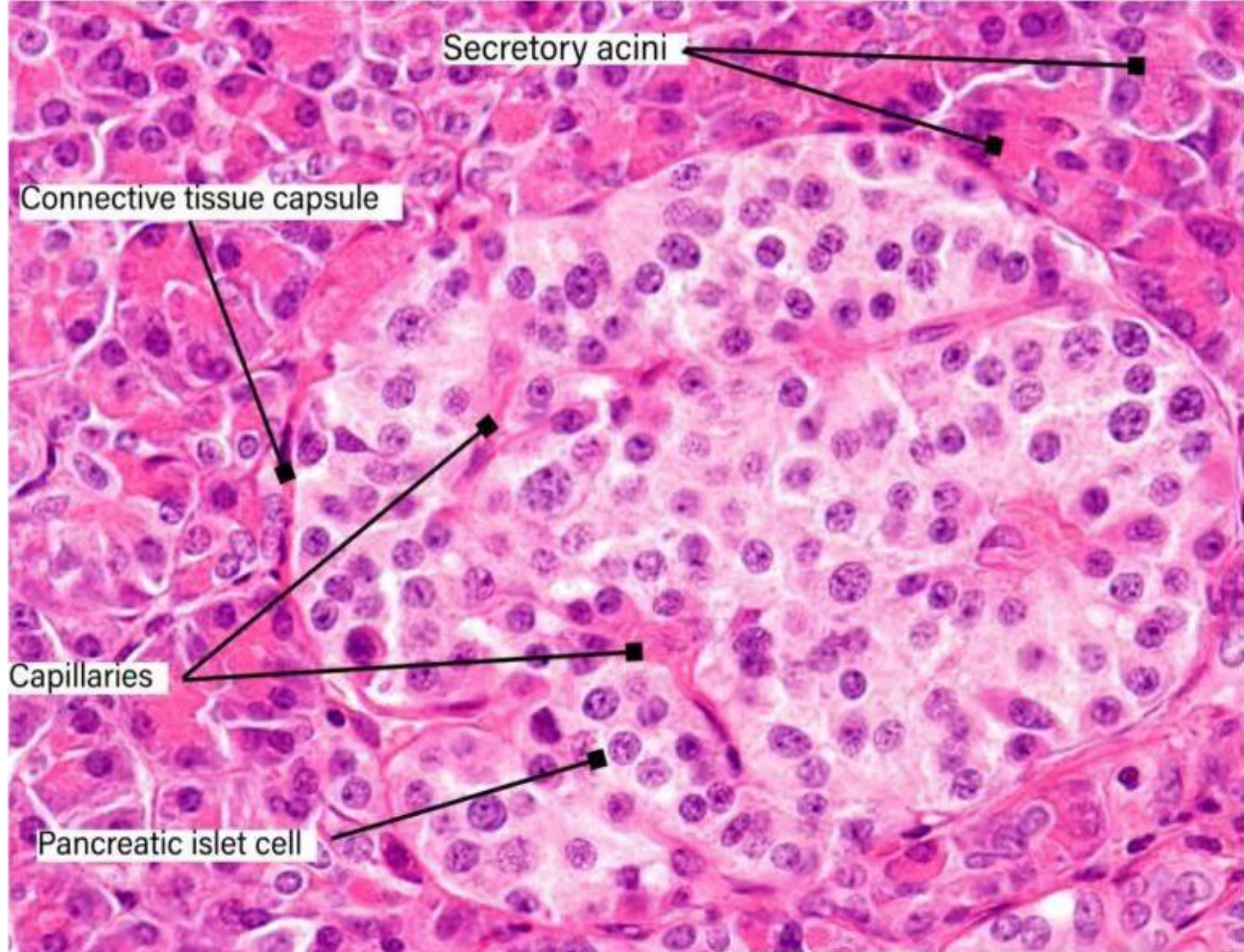




### Islet of Langerhans









THANKS